

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

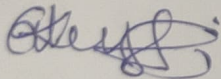
Scenario-Based

Code of Conduct:

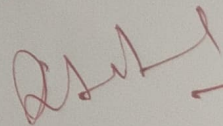
I Gk. vyshnavi / 192425069 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



19/20



1. Scenario:

A smart campus has been allocated the IP block 192.168.100.0/24. Different departments require different subnet sizes.

Department	Subnet Mask
School of Computing	/25
School of Electronics	/26
School of Mechanical	/27

Step 1:

1. School of Computing.

Subnet mask: 255.255.255.128

Block size: 128

Network address: 192.168.100.0

Broadcast address: 192.168.100.127

Valid host range: 192.168.100.1 - 192.168.100.126

Usable Hosts:

$$2^{(32-25)} - 2 = 2^7 - 2 = 126$$

2. School of Electronics

Subnet mask: 255.255.255.192

Block size: 64

Network addresses: 192.168.100.128

Broadcast address: 192.168.100.191

Host Range: 192.168.100.191 - 192.168.100.190

Usable hosts:

$$2^6 - 2 = 62$$

3. school of mechanical engineering:

Subnet mask: 255.255.255.224

Block size: 32

network addresses: 192.168.100.192

Broadcast address: 192.168.100.223

Host Range: 192.168.100.193 - 192.168.100.222

usable Hosts: $2^5 - 2 = 30$

4. Remaining unused IP Range:

192.168.100 - 192.168.100.255

allocated:

1. 192.168.100.0 - 127

2. 192.168.100.128 - 191

3. 192.168.100.192 - 223

Remaining capacity: 32 Addresses.

2.

Scenario

Company network:

10.10.0.0/6

1. Software development (124)

Subnet mask

255.255.255.0

network address: 10.10.0.0

Broadcast address: 10.10.0.255

Host range: 10.10.0.1 - 10.10.0.254

usable host: $2^8 - 2 = 254$

2. Quality Assurance:

network address: 10.10.1.0

Broadcast: 10.10.1.127

Host range: 10.10.1.1 - 10.10.1.126

3. Human Resources.

network address - 10.10.1.128
Broadcast - 10.10.1.191
Host range - 10.10.1.129 - 10.10.1.190
usable Hosts - 62.

4 Executive management

Network address:
10.10.1.192
Broadcast address: 10.10.1.223
Host Range: 10.10.1.193 - 10.10.1.222
usable Hosts: 30

IPV6 Address

2001:0db8:abcd:0000:0000:0000:0000:/64

1. compressed form.

leading zeros removed

consecutive zero blocks replaced by: 2001:db8:abcd::/64

2. Expanded form:

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network prefix

Prefix length: /64

network prefix: 2001:db8:abcd:0000::/64

Host portion: last 64 bits.

4. no. of host address

Remaining bits: $128 - 64 = 64$

5. Total possible hosts: IPv6 has no broadcast address.

$2^{64} = 18,446,744,073,709,551,616$

4. Routing table:

Destination network:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Destination IP:

192.168.2.45

1. Destination subnet

192.168.2.45 belongs to 192.168.2.0/24

2 matching Routing entry:

192.168.2.0/24

3. Next hop:

packet is forwarding through the interface connected to 192.168.2.0/24

4. Default route:

If no entry matches:

Default route

0.0.0.0

Router forwards the packet to configured default gate way.

5. Administration LAN

192.168.10.0/26

Finance LAN

192.168.10.64/26

Gate ways: administration: 192.168.10.1

Finance: 192.168.10.65

Administration lan:

network:

192.168.10.0

Broadcast: 192.168.10.63

Host Range: 192.168.10.1 - 192.168.10.62

PC1 → 192.168.10.2

PC2 → 192.168.10.3

PC3 → 192.168.10.4

Gateway: 192.168.10.1

2. Finance lan

Network : 192.168.10.64

Broadcast: 192.168.10.127

Host range: 192.168.10.65 - 192.168.10.126

~~PC1 → 192.168.10.66~~

~~PC2 → 192.168.10.67~~

PC3 → 192.168.10.68.